Example

Problem

- Find the exact total cost (running time) of the following recurrence equation using recursion tree.
  - Let's assume that n is the exact power of 2.

$$T(n) = \begin{cases} 1 & , n = 1 \\ T\left(\frac{n}{2}\right) + n^3 & , n > 1 \end{cases}$$

level : 0 $\quad n^3$ ← cost of node.

level : 1 $\quad \left(\frac{n}{2}\right)^3$

level : 2 $\quad \left(\frac{n}{4}\right)^3$

⋮

level $\log n$ $\quad 1$

$\frac{n}{2^i} = 1 \Rightarrow i = \log n$

$\therefore$ # of level : $(\log n + 1)$

$$\sum_{i=0}^{\log n} \left(\frac{n}{2^i}\right)^3 = n^3 \sum_{i=0}^{\log n} \left(\frac{1}{8}\right)^i = n^3 \cdot \frac{\left(\frac{1}{8}\right)^{\log n + 1} - 1}{\frac{1}{8} - 1}$$

$$= -\frac{8}{7} n^3 \cdot \left(\left(\frac{1}{8}\right)^{\log n + 1} - 1 \right)$$

$$= -\frac{8}{7} n^3 \left\{ \left(\frac{1}{8}\right)^{\log n} \cdot \left(\frac{1}{8}\right) - 1 \right\}$$

$$= -\frac{8}{7} n^3 \left\{ n^{\log 2^{-3}} \cdot \left(\frac{1}{8}\right) - 1 \right\}$$

$$= -\frac{1}{7} + \frac{8}{7} n^3 = \frac{8}{7} n^3 - \frac{1}{7}$$